

Lecture 01:

Introduction

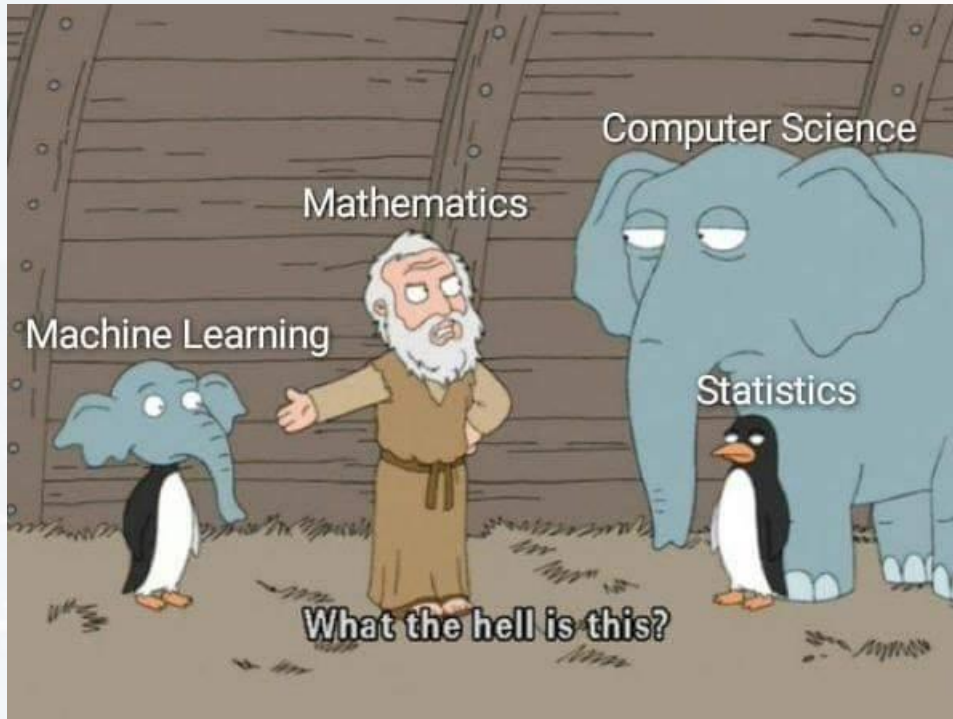


Markus Hohle

University of California, Berkeley

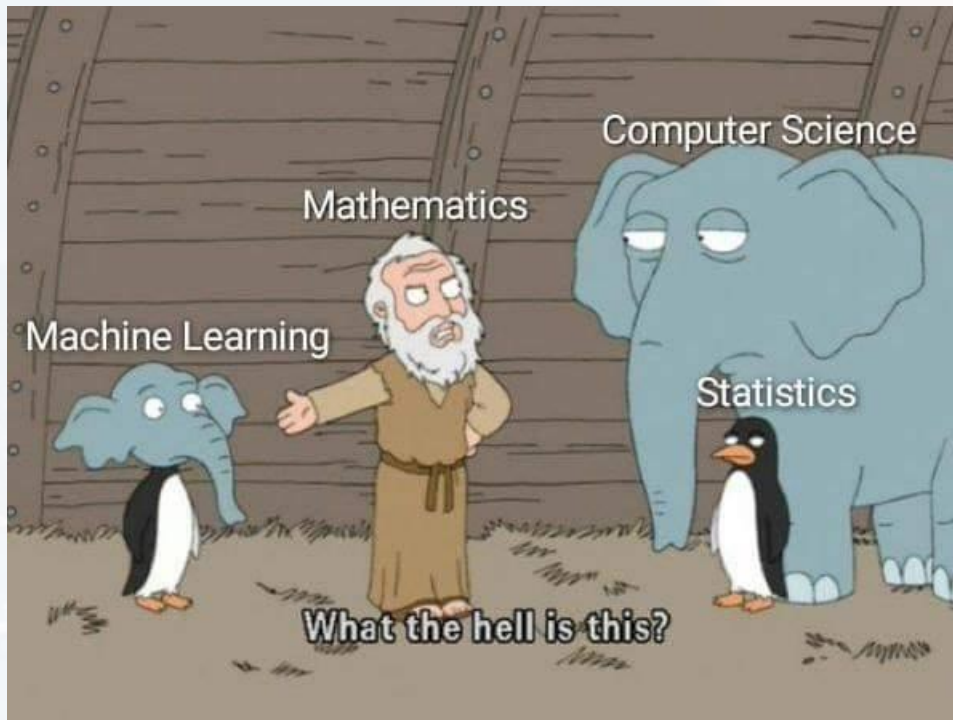
Machine Learning Algorithms

MSSE 277B, 3 Units



Outline

- Motivation
- What is Machine Learning?
- The Lecture
- Course Map



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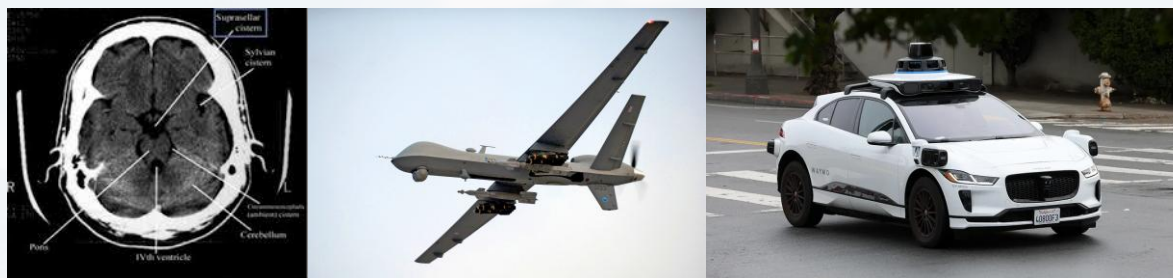
Why Machine Learning?

- **automating workflows**

- face recognition, identification/authentication tools

- **manufacturing processes, quality control**

- **repetitive, standardized workflows**



- **handling large data sets**

- **epigenetic data for drug development**

- traffic management

- **disease control**

- stock market dynamics

- **generating new data**

- **new materials (semi/superconductor, linker proteins etc)**

- AI generated images/videos

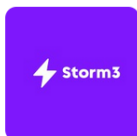


Why Machine Learning?

LinkedIn: 1st three Job Postings, Dec 19th 2025

Top job picks for you

Based on your profile, preferences, and activity like applies, searches, and saves



Research Scientist - Vision Data Infrastructure

Storm3 • San Francisco, CA (Hybrid) • \$250K/yr - \$600K/yr • Vision benefit



Actively reviewing applicants

Promoted • Easy Apply



Staff AI Engineer

Lumicity • San Francisco Bay Area (Hybrid) • \$250K/yr - \$280K/yr • 4 benefits



Actively reviewing applicants

Promoted • Easy Apply



Staff AI/ML Engineer

Harnham • Sunnyvale, CA (On-site) • \$200K/yr - \$225K/yr • 4 benefits



Actively reviewing applicants

Promoted • Easy Apply



What You Bring

- 5+ years building and deploying **ML/AI systems** at scale.
- Strong **Python/Java/C++** skills and experience with **ML frameworks** (TensorFlow, **PyTorch**).

Requirements:

- Strong experience with **data engineering, computer vision, or machine learning infrastructure**.
- Expertise in building and scaling **ETL/data pipelines** for large unstructured datasets.
- **Proficiency with Python, PyTorch,** and distributed data frameworks (e.g., Ray, Spark, **Dask**).
- Experience with **WebDataset, TFRecords, Parquet**, or similar high-throughput data formats.
- Familiarity with **GPU-accelerated** preprocessing, **NVIDIA DALI**, or equivalent systems.



Actively reviewing applicants



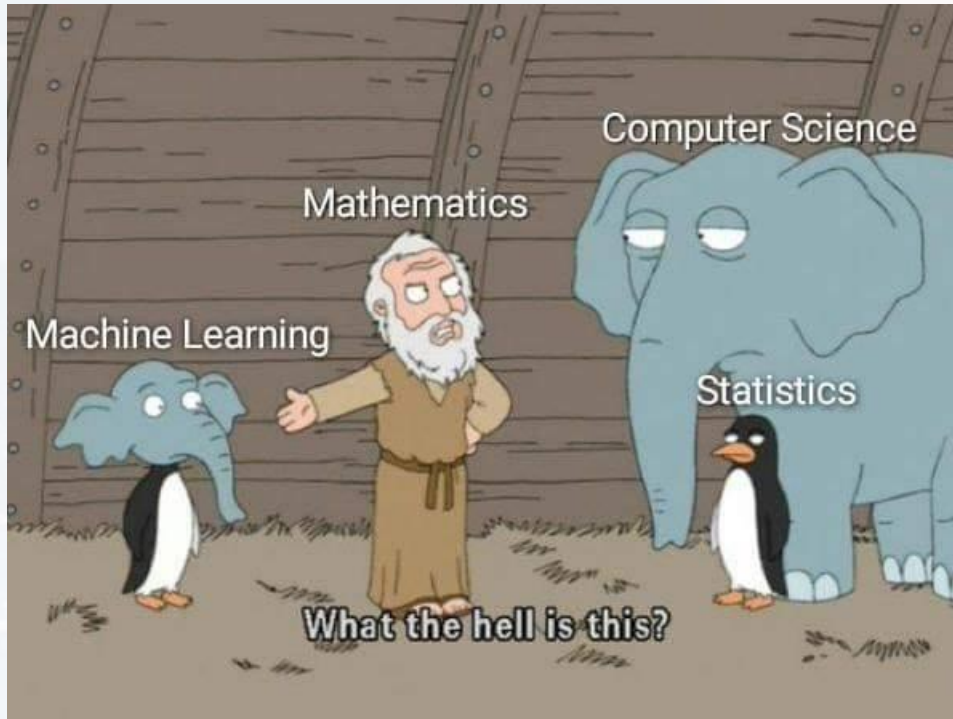
Why Machine Learning?

*Listen up son, there are three languages you need to learn
in order to become successful:*

- English
- Math
- Python *)



*) or C++



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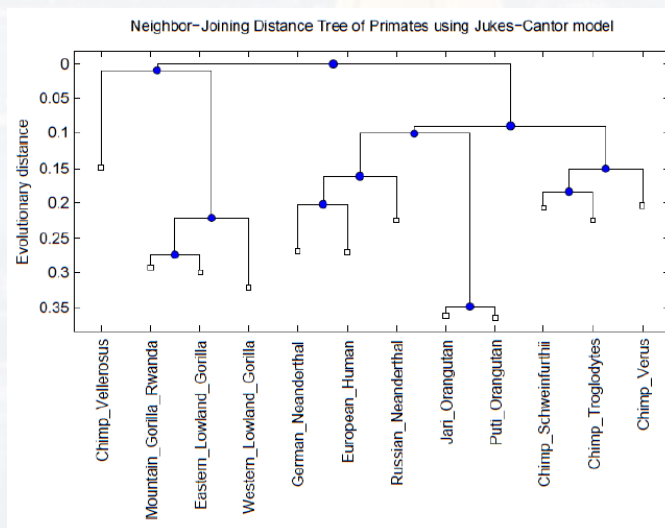
Wiki says:

“Machine learning (ML) is a field of study in *artificial intelligence* concerned with the development and study of *statistical algorithms* that can learn from data and generalize to unseen data, and thus perform tasks *without explicit instructions*”

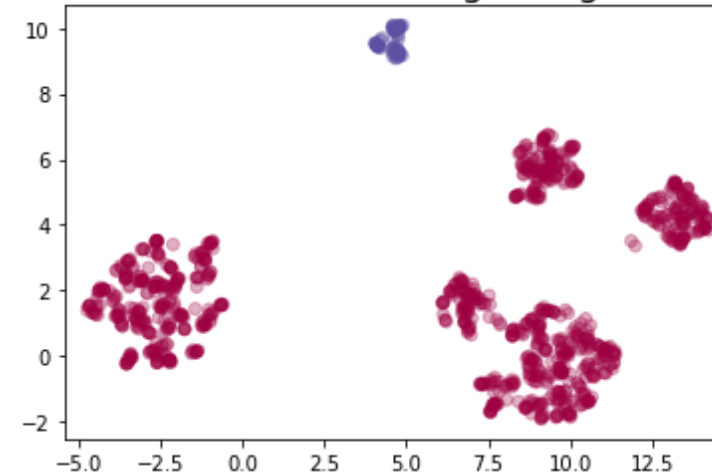
→ more flexible than standard programs

- artificial intelligence
- statistics

cluster and/or categorize data and understand relations/interactions, find pattern

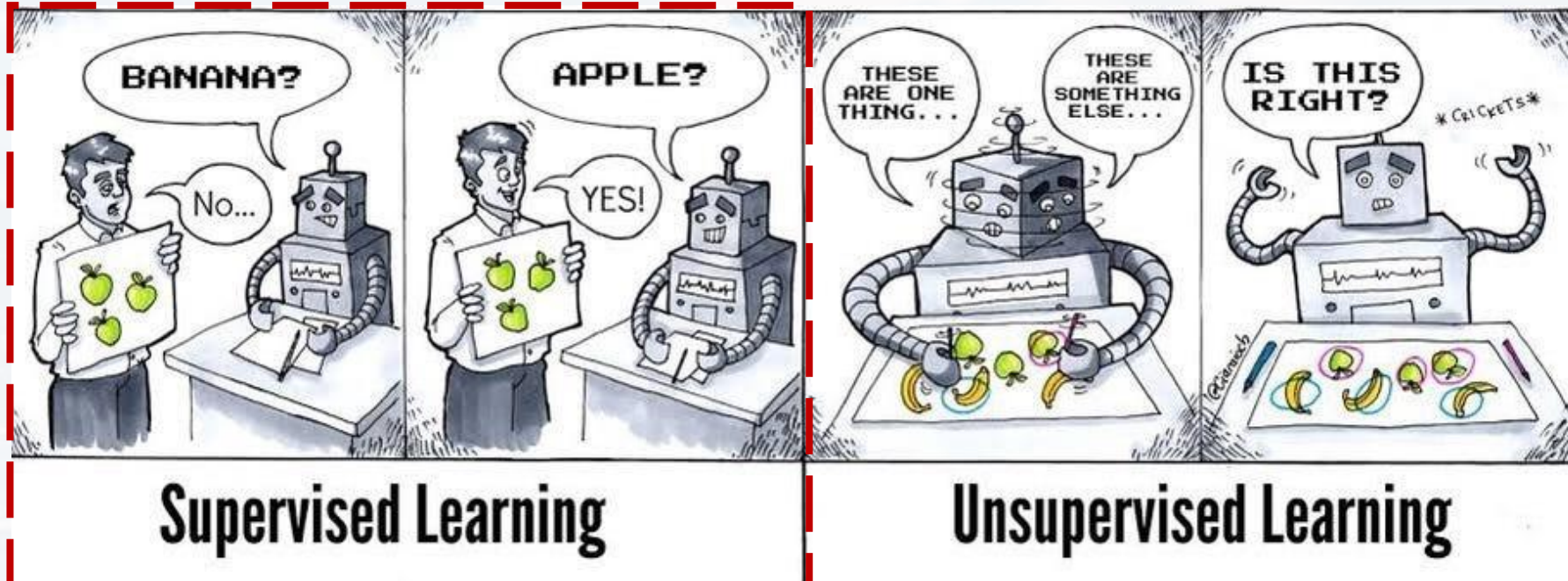


UMAP projection of the AD dataset (age range(0: <75; 1: >=75)_1)





curve fitting, linear/logistic models:
training data set and a **test** data set...

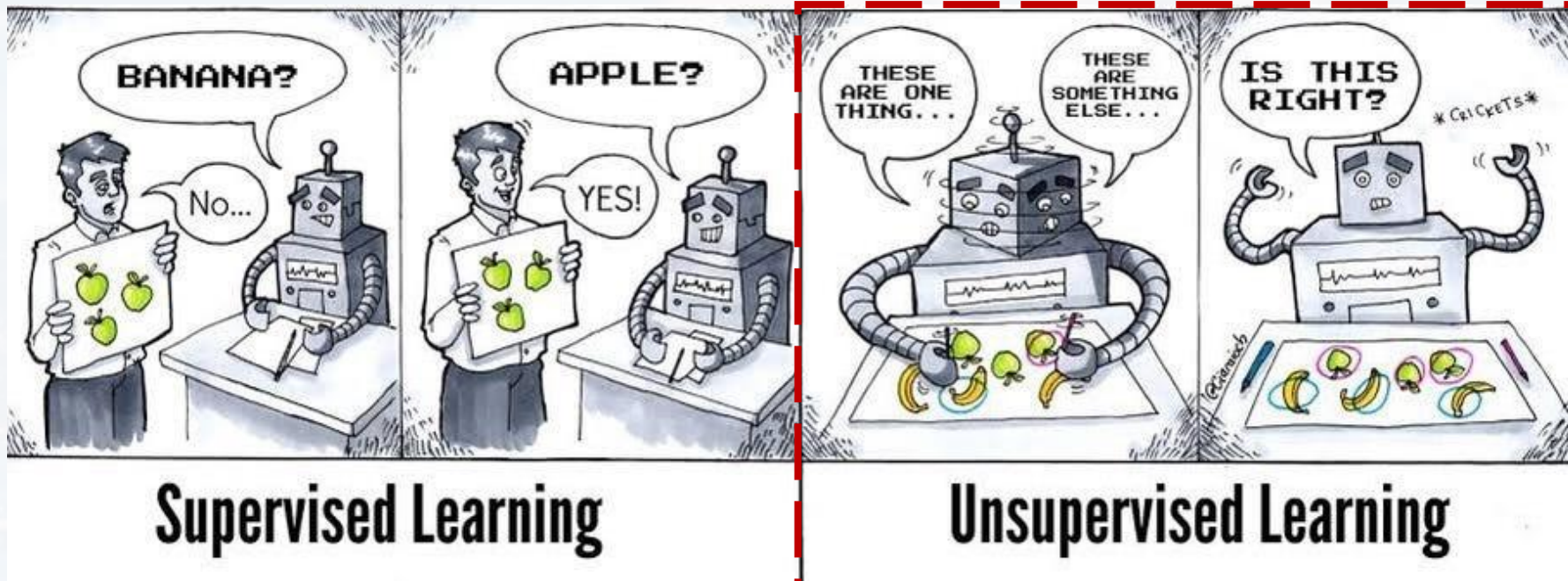


- Support Vector Machine
- K-nearest
- ANNs



curve fitting, linear/logistic models:
training data set and a **test** data set...

no training data set required – we can start right away!

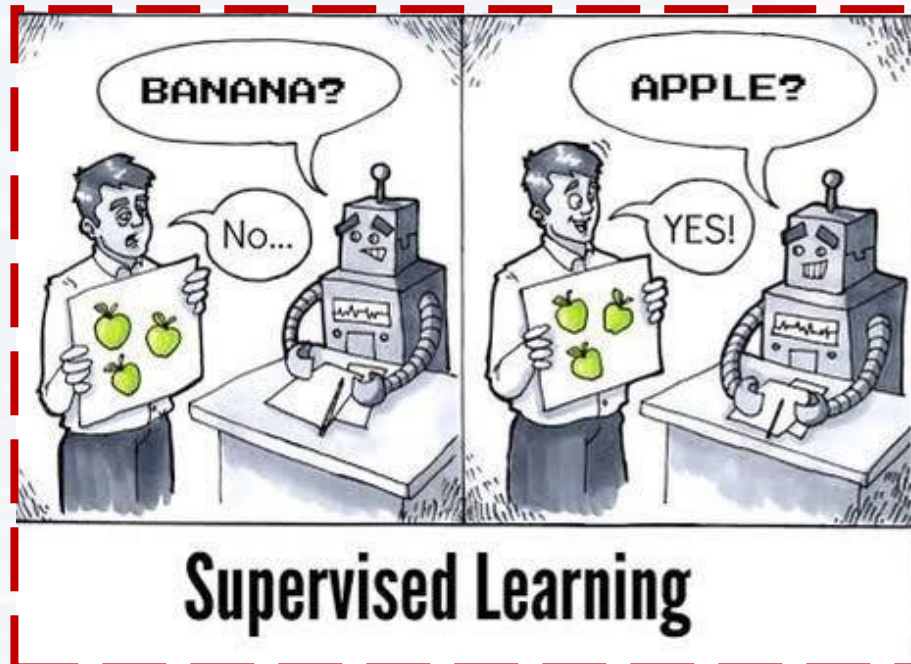


- Support Vector Machine
- K-nearest
- ANNs

- K - means
- GMM
- trees



curve fitting, linear/logistic models:
training data set and a **test** data set...



- Support Vector Machine
- K-nearest
- ANNs

1) creating the model:

```
my_model = library.method(argument1 = 'arg1', ... )
```

2) training the model

```
out = my_model.fit(xtrain, ytrain)
```

3) evaluation

```
ypred = out.predict(xeval)  
accur = (ypred == yeval).sum()/len(yeval)
```

4) prediction (actual application)

```
ypred = out.predict(xnew)
```



no training data set required – we can start right away!

1) creating the model:

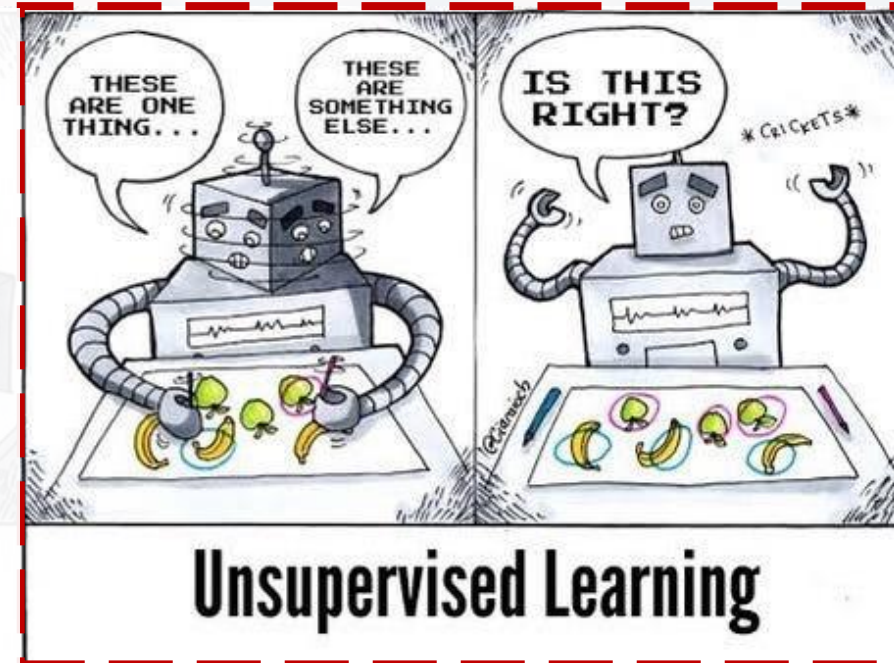
```
my_model = library.method(argument1 = 'arg1', ... )
```

3) evaluation (*if* y is known)

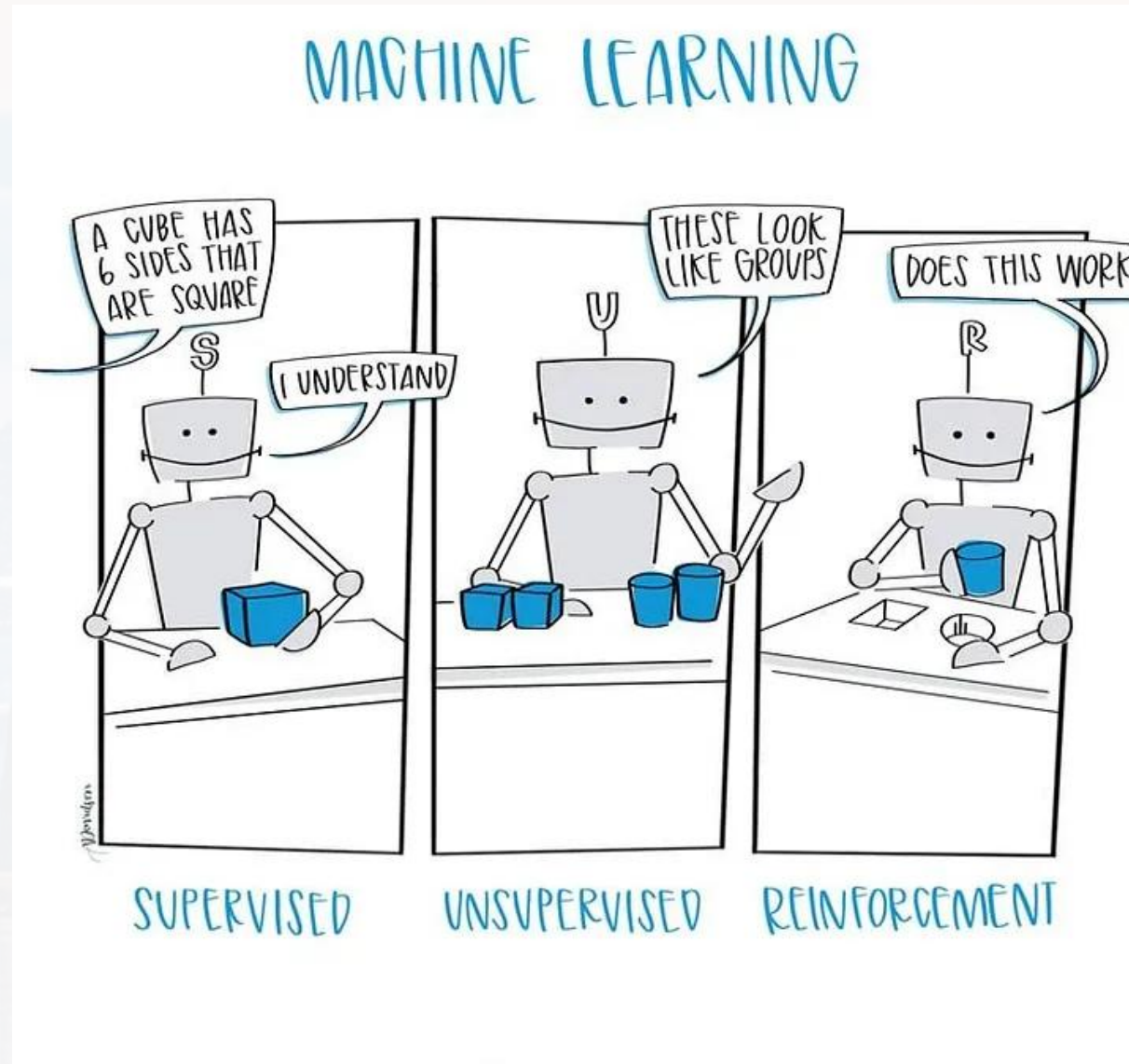
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4) prediction (actual application)

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ypred = out.predict(xnew)
```



- K - means
- GMM
- trees

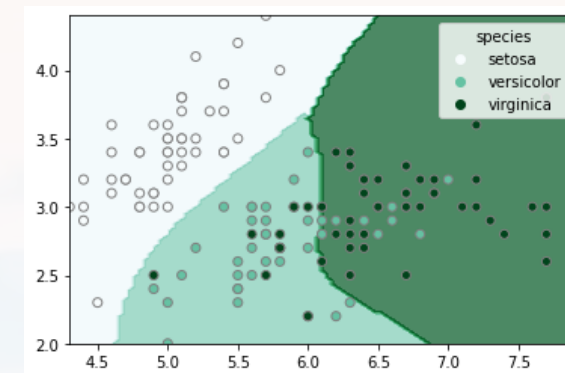




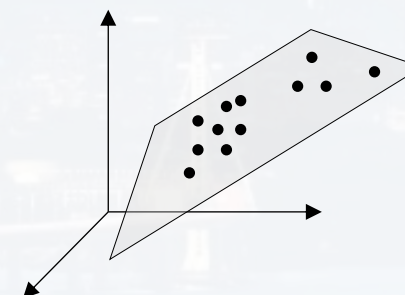
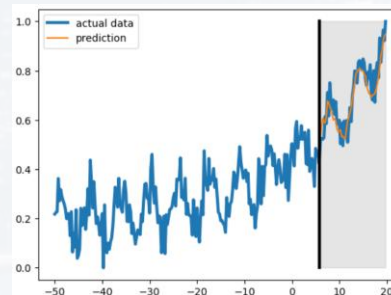
tasks:

optimizing an
objective
function

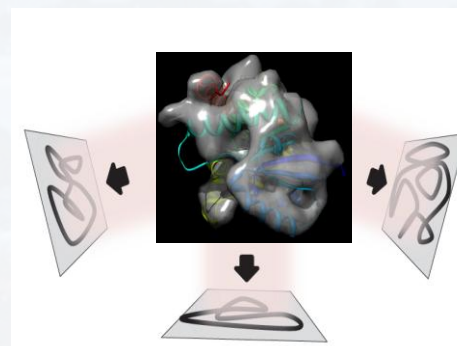
- classification



- regression



- generation





tasks:

- classification

Index	molecular_weight	electronegativity	bond_lengths	num_hydrogen_bonds	logP	label
0	341.704	2.65585	3.09407	2	9.11147	Toxic
1	335.951	3.22262	2.89039	7	8.92848	Toxic
2	235.203	2.44115	2.48203	1	6.49731	Non-Toxic
3	246.505	2.76656	2.71547	7	7.45089	Non-Toxic
4	437.939	3.4801	3.59569	3	10.9156	Non-Toxic

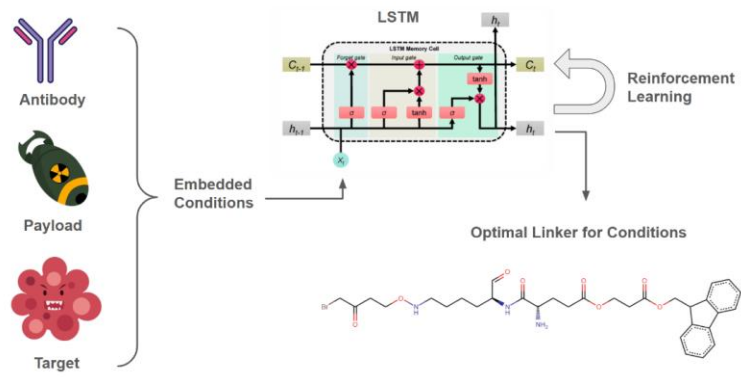
- regression

Index	molecular_weight	electronegativity	bond_lengths	num_hydrogen_bonds	logP	toxicity_score
0	341.704	2.65585	3.09407	2	9.11147	80.9281
1	335.951	3.22262	2.89039	7	8.92848	83.4911
2	235.203	2.44115	2.48203	1	6.49731	61.8406
3	246.505	2.76656	2.71547	7	7.45089	57.0538
4	437.939	3.4801	3.59569	3	10.9156	131.326

Final Project 277B Fall 2025

Optimizing ADC Linkers via LSTM Reinforcement Learning

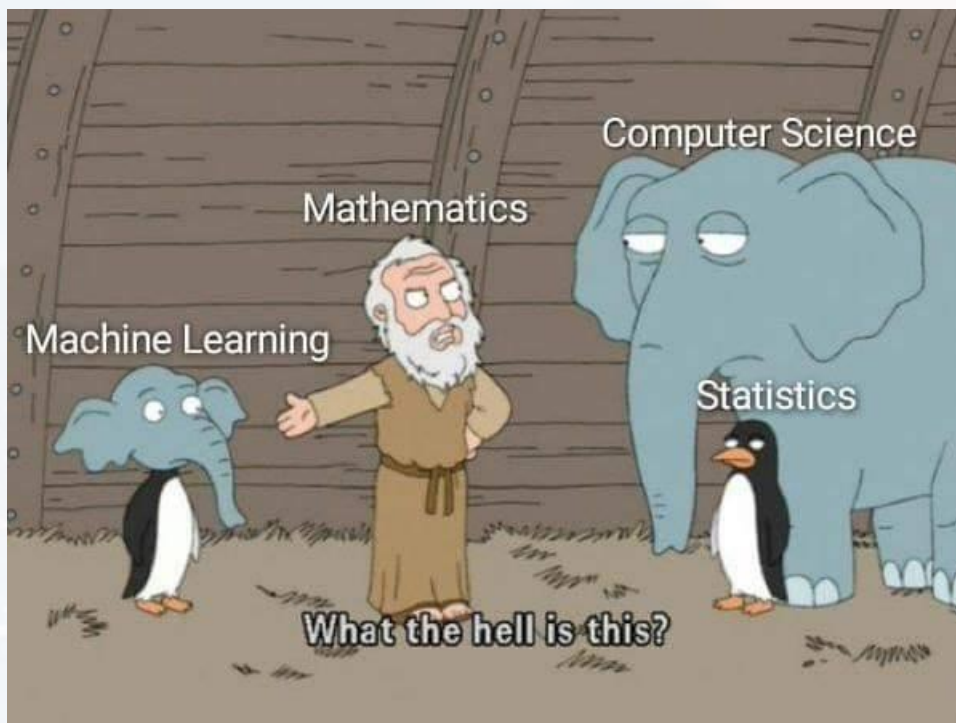
Alex Chase, Jonas Kazimli, Sana Khan, Hailey Monaco, Natalia Rivera, and Leah Sutherland



- generation

“create a molecule with the properties XYZ,
such that it is non-toxic”

carrier or linker molecules (drug development)



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GSI:

Spencer Uyematsu

Bioinformatics Scientist at Proteios

Technology

MSSE alumnus



Lecturer:

Markus Hohle

Lecturer at UC Berkeley &

Data Analysis Consultant

PhD Physics





Lecture: Mo 3:00 – 4:00pm PT

Discussion (Spencer/Markus): Th 1:00 – 3:00pm PT

Office Hours (Markus): Th 4:00 – 6:00pm PT

- **you can pick a ML/AI project**
- **work in teams of 4/5**
- **start planning after 1/3 of the course**
- **more detailed intro after ½ of the course**
- **submit proposal**
- **checkpoint**
- **presentation**

Material:

pre recorded Lectures: → bcourses

codes/slides → bcourses & GitHub

HW assignments → bcourses

Grades:

HW assignments: → 50%

Final Project → 40%

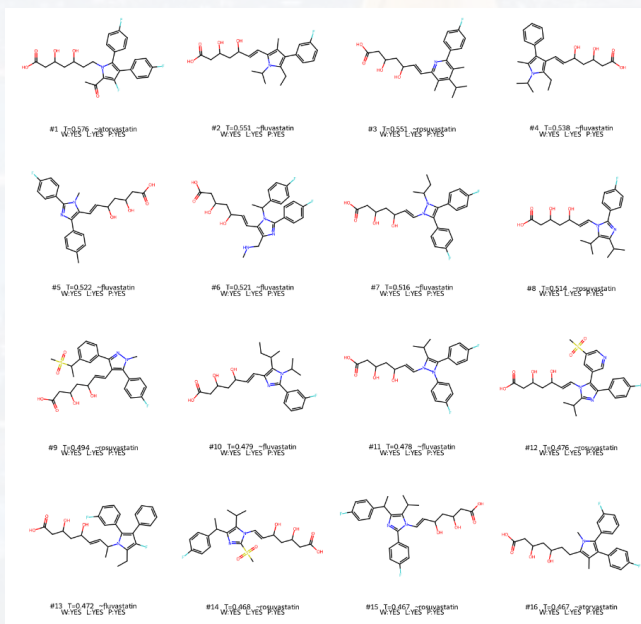
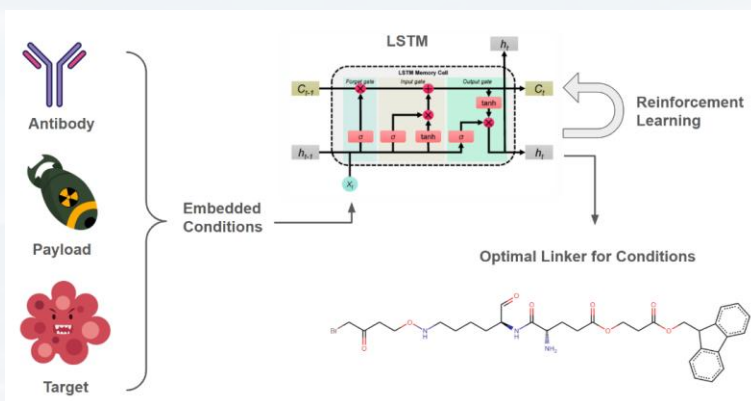
Course Attendance → 10%



Final Project examples

Optimizing ADC Linkers via LSTM Reinforcement Learning

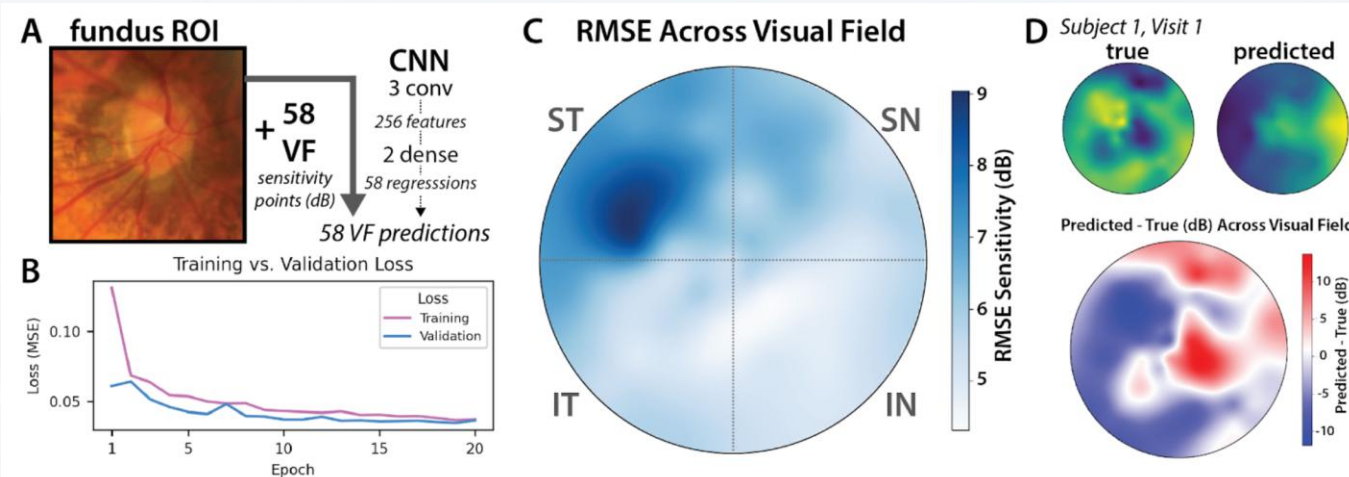
Alex Chase, Jonas Kazimli, Sana Khan, Hailey Monaco, Natalia Rivera, and Leah Sutherland



Machine Learning Applications for Visual Field Analysis: A Study of Vision Loss Progression

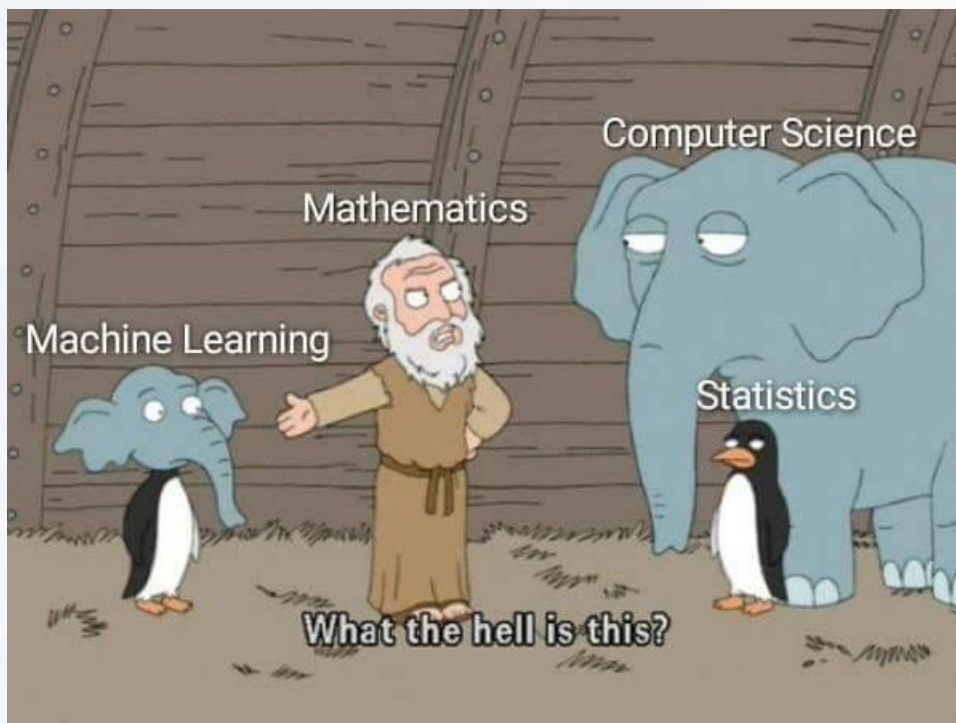
Authors: David Houshang, Lily Hirano, Kirk Ehmsen, Yash Maheshwaran, Christian Fernandez

University of California at Berkeley - College of Chemistry - Team 1



Evidence-Aware Validation of AI-Generated Drug Candidates Using PPI Networks and Network Pharmacology

D. Kim, Z. X. Rothering, E. Gonzalez de la Vega Arteaga,
D. D. Edwards, S. Tijare, G. Krishna



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Lecture 1: Course Overview and Introduction to Machine Learning

Lecture 2: Bayesian Methods in Machine Learning

classic ML tools & algorithms

Lecture 3: Dimensionality Reduction: Principal Component Analysis

Lecture 4: Linear and Non-linear Regression and Classification

Lecture 5: Unsupervised Learning: Clustering and Gaussian Mixture Models

Lecture 6: Adaptive Learning and Gradient Descent Optimization Algorithms

Lecture 7: Introduction to Artificial Neural Networks - The Perceptron

ANNs/AI/Deep Learning

Lecture 8: Introduction to Artificial Neural Networks - Building Multiple Dense Layers

Lecture 9: Convolutional Neural Networks (CNNs) - Part I

Lecture 10: CNNs - Part II

Lecture 11: Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTMs)

Lecture 12: Combining LSTMs and CNNs

Lecture 13: Optional: Running Models on GPUs and Parallel Processing

Lecture 14: Project Presentations

Lecture 15: Transformer

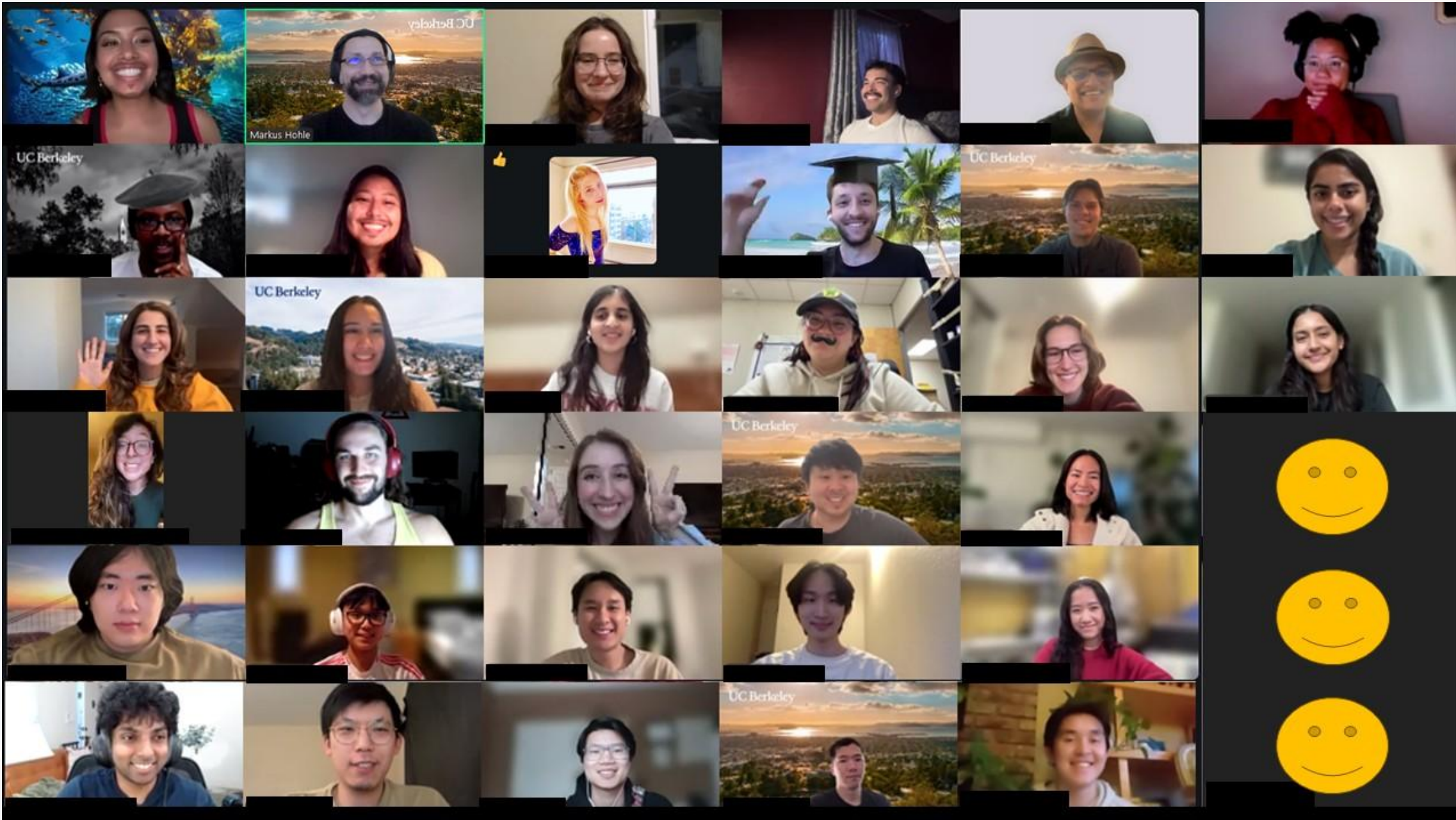
Lecture 16: GNN



Introduction

Course Map

May 2026



Thank you very much for your attention!

?Questions?